

House Price Prediction in King County Explained Through Stacking Ensemble and Ablation Interpretation

Yuyang Zhang

School of Computer Science
University of Nottingham Ningbo China
Ningbo, China
Student ID: 20514470
scyyz26@nottingham.edu.cn

Abstract—Predicting house prices is one of the main tasks in real estate research. However, house price data are characterized by non-linearity, spatial dependency, and heavy tails; moreover, no model has been found to perform better than others in all cases. Stacking ensembles can be used to combine complementary algorithms. Still, the approach to predicting house prices is inconsistent in the literature, and most studies do not reveal the contribution of each base learner when building an ensemble. In this paper, an end-to-end pipeline is developed on the King County housing data (4,600 observations, 18 variables) to answer the following questions: does stacking yield better results compared to the best-performing individual model, and which base learners have the most influence on performance improvements? To begin with, six base learners are evaluated against the same preprocessing procedure: Ridge, Random Forest, Extra Trees, GBR, XGBoost, and LightGBM. Then, three stacking schemes are built on the same base learners using out-of-fold predictions and either Ridge or LassoCV as the meta-model. The best stacking configuration reaches $R^2 = 0.8015$ on the held-out test set, representing a gain of 0.025 over the strongest individual model LightGBM ($R^2 = 0.7761$). A leave-one-out ablation shows that LightGBM is the dominant contributor inside the ensemble ($\Delta R^2 = 0.0241$), with Ridge as the next most valuable component ($\Delta R^2 = 0.0177$) despite being the weakest standalone learner. The remaining four base learners contribute between 0.0139 and 0.0153. Therefore, increasing the number of base learners is not necessarily an effective way to improve ensemble performance since it is mostly determined by several strong learners and spatial features encoded as target means.

Index Terms—house price prediction, stacking ensemble, ablation study, target encoding, gradient boosting

I. INTRODUCTION

House prices are a crucial concern for buyers, investors, agents, and policy makers [1], and they are notoriously difficult to model. Each property constitutes an almost unique combination of structures, location, and condition, where the relationship between each factor and price is nonlinear and imbalanced across the market [2]. Methods based on additive linear relationships, like hedonic regression and other parametric models, struggle to account for the complex interactions and give inconsistent prices when transaction numbers are sparse [3].

Machine learning techniques relax these assumptions and are now ubiquitous methods for property valuation. Nevertheless, none of these methods performs consistently across all datasets and markets. Ridge, linear models in general, are easy to interpret but incapable of modeling non-linearities [4]. On the other hand, tree-based ensembles perform well in modeling interactions but suffer from over-fitting and interpretability issues [5], [6]. Deep networks are highly flexible but architecture choices and tuning affect results [7], [8]. This problem motivates stacking, where a meta-model combines the outputs of several heterogeneous learners so as to keep the best of each one. Yet, stacking is not extensively used in real estate price prediction due to inconsistent advantages reported so far [9]. Most of the related literature uses single learners and simple combinations of learners, such as bagging and boosting algorithms [6]. The few studies that use stacking show some gains but treat the ensemble as a black-box approach with little discussion of what each of the base learners contributes to the prediction task [10]. Without this understanding, it is uncertain if the improvements generalize or a simpler ensemble would work just as well.

This study tried to fill the gap using the King County housing dataset (4,600 transactions, 18 features, May 2014 to May 2015) [11]. In particular, it combines structural attributes (number of bedrooms, bathrooms, and square feet of living space), environmental characteristics (waterfront and view), and high cardinality location features (city and state/zip code), which allows various learners to take advantage of each piece of information differently. Contributions are:

- A controlled comparison of six baseline regressors and three stacking strategies with identical preprocessing.
- A leave-one-out experiment analyzing how important each base learner is to the stacked model through its contribution to the final prediction, aided by meta-learner coefficients.
- Identify that the LightGBM model is the best among the models used in the ensemble, whereas the model with the least contribution is the Ridge model, which is the least

accurate individually.

II. LITERATURE REVIEW

Data-driven approaches to estimating the value of houses can be classified into three broad groups: parametric baselines, single-learner machine learning, and ensemble models.

For parametric baseline models, Hedonic regression models assume a linear relationship between the price and characteristics of a property [12]. They are straightforward and cost-effective to apply, but due to their linear assumption, they are susceptible to non-linearity and spatial heterogeneity in the data [4], [13].

As machine learning gained traction in this area, a range of single-learner models has been applied to housing valuation. Tree-based learners, decision trees [14], random forests [15], and gradient boosting [16], [17] — are the most widely used, because they do not require feature scaling and capture feature interactions without explicit transformation of the target. Neural networks have outperformed linear baselines in some comparisons [13] but are prone to overfitting and offer limited interpretability [4]. Support vector regression with a non-linear kernel has also been applied, but is sensitive to the choice of C and the ϵ -insensitive loss [18]. The recurring conclusion across these studies is that any of these learners can be made to work well on a given dataset with careful tuning, but no single family is consistently best.

This is the gap that ensemble methods are designed to close. Bagging and boosting reduce the variance of individual learners by aggregating many of them [15], [16]. Stacked generalisation goes further by training a meta-learner on the predictions of heterogeneous base models [19], and is in principle better placed than a single ensemble family to exploit complementary errors. In the housing-valuation literature, however, stacking remains relatively rare. Existing studies on stacking in housing valuation primarily focus on overall predictive accuracy, with improvements typically marginal compared with well-tuned individual baseline models [9], [10], and rarely decompose the ensemble to show which base learners actually contribute. Bjørgve et al. [10] note this explicitly, arguing that the practical case for stacking in housing valuation is still weak because gains are often modest and the underlying mechanism is opaque.

III. METHODOLOGY

The pipeline has six stages: exploratory analysis, cleaning, feature engineering, feature selection, baseline modelling, and stacking. Each stage is described below, with leakage-aware design choices flagged where they apply.

A. Exploratory Analysis

The dataset is loaded from `data.csv` and inspected for missing values, dtype consistency, and target skew. No imputation is needed. The price distribution is right-skewed with a long upper tail, so outlier handling is required before training error-sensitive metrics such as RMSE. Continuous

predictors are visualised against price (regression plots), categorical predictors via boxplots, and pairwise relationships via a correlation heatmap.

B. Outlier Removal

Two stages of data cleaning have been performed. The first stage removes 49 samples with zero price because such values are erroneous data entries and would be mapped to $\log(1) = 0$ following the log transformation stage. The second stage involves the removal of samples above the 99th percentile of price (46 samples, 1.0% of the data).

C. Target Transformation

House prices are heavily right-skewed with the 99th-percentile filter. In order to stabilise training, the target is log-transformed using $\log(1 + y)$ before fitting any model, and predictions are inverted with $\exp(y) - 1$ when calculating R^2 , RMSE, and MAE.

D. Feature Engineering

The `date` column is used to derive temporal features, including `house_age`, `is_renovated`, and `years_since_renovation`, which capture the age and renovation history of the property. The feature `basement_ratio` represents the relative size of the basement, while `has_basement` is a binary indicator of basement presence. A small constant is added to avoid division by zero when computing the ratio. The feature `bath_bed_ratio` contains information on the number of bathrooms and bedrooms, providing additional structural information. The `month` variable is converted into seasonal categories and subsequently encoded using one-hot encoding.

E. Encoding and Train-Test Split

After the application of feature engineering techniques, raw identifiers (`street`, `city`, and `statezip`) are dropped. The data are split into training (80%, 3,643 instances) and test (20%, 911 instances) sets using a random split with a fixed seed for reproducibility. To avoid the dimensional growth and at the same time preserve location information, the categorical variables `city` and `statezip` are reintroduced using target encoding. Each category is replaced by the mean house price computed from the training data. For categories in the test set that are not observed during training, the global mean of the training set is used. All encoding statistics are computed exclusively on the training data to prevent data leakage.

F. Feature Selection and Scaling

The feature selection step combines automatic and manual techniques. Features are selected automatically based on the criteria of correlation with price: $|r| > 0.05$. In addition, a manual set of domain-relevant features is saved regardless of their univariate correlation. Features with high mutual correlation, $|r| > 0.85$, among other retained features are pruned. Two exceptions from this rule are `sqft_living`, `bathrooms`, and the target-encoded location columns. As a result of this operation, only 14 features are left (dropped is `basement_ratio`). Remaining features are standardized based on training data only.

G. Baseline Models

Six baseline models belong to three groups. The ridge regression model is used as the baseline for regularized linear models. The Random Forest and Extra Trees are two bagging models. GBR, XGBoost, and LightGBM are three boosting models, where LightGBM follows a leaf-wise tree growth method, which usually works well with XGBoost’s depth-wise approach. There was also consideration of using an MLP at first; however, due to its instability in the log-target problem, it had to be removed from the ensemble.

H. Stacking Ensemble

Out-of-fold (OOF) predictions are generated for each base learner using 5-fold cross-validation. For each fold, predictions on the held-out subset are used to construct meta-features, while test predictions are averaged across folds. Three stacking configurations are then trained on these meta-features. Configuration (V1) uses only the OOF predictions, utilizes a Ridge meta-learner. In the second configuration (V2), the meta-features are concatenated with the original, scaled features before feeding into the Ridge meta-learner. Lastly, the third configuration (V3) substitutes LassoCV for the meta-learner, thus introducing automatic regularization and feature selection.

I. Evaluation

All models are assessed using the held-out 20% test split. To reach the second research goal, a leave-one-out ablation on the top-performing stacking approach is performed, whereby the impact of each base learner on the test R^2 is measured by removing one base learner at a time. Additionally, meta-learner coefficients for V1 and V3 are provided for two reasons. First, they show how the predictions from different base learners are combined and indicate any redundancy in the models through correlation of the coefficients. Second, for configurations what including Ridge meta-learner, V2 outperforms V1 (See details in Results). Furthermore, Random Forest feature importances and three diagnostic plots (predicted-versus-actual, residuals against predictions, residual histogram) are used to inspect bias and heteroscedasticity.

IV. EXPERIMENTAL RESULTS

The results will be presented in the same order as the pipeline above: observations from EDA, followed by an evaluation of all models, an ablation study, meta-learner coefficients, and diagnostics.

A. EDA and Feature Engineering Insights

The price distribution is an indication of the necessity of dealing with outliers. The price distribution shown in Fig. 1a is extremely skewed to the right, where most data points are clustered on the left side while several points are scattered far out to the right. This is presented graphically using a box plot in Fig. 1b. The price distribution in Fig. 1c is compressed after removing the top 1% of the data points. After applying a log transformation to the price, the skewness has been reduced. (Fig. 1d)

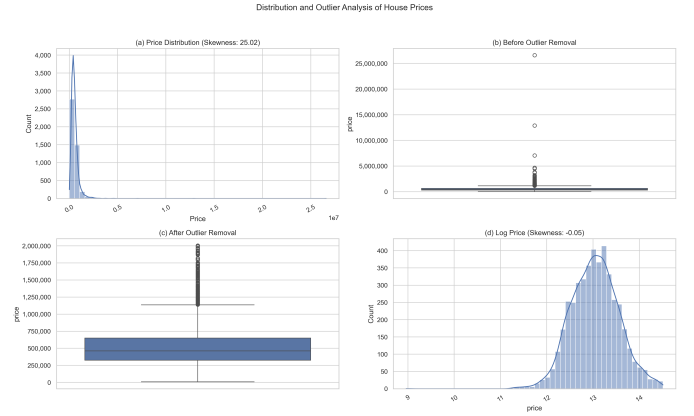


Fig. 1: Price distribution and outlier treatment. (a) Original distribution (right-skewed). (b) Before outlier removal. (c) After removing the top 1% of values. (d) Log-transformed price distribution.

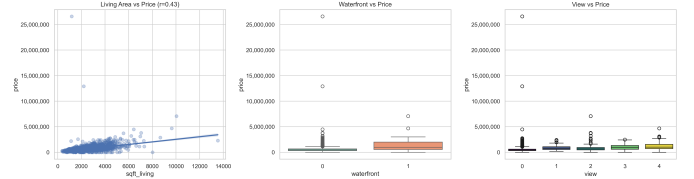


Fig. 2: Key predictors against price. (a) Living area is positively correlated with price. (b) Waterfront properties command a clear premium. (c) Higher view ratings track higher prices monotonically.

Fig.2 depicts the three main factors that guided our feature engineering efforts. Panel (a) shows a plot between $sqft_living$ and price, showing that there is an increasing trend, as well as increasing variation for higher values of this feature, implying a non-linear relationship. Panel (b) indicates that properties that have waterfronts form a group with higher prices. Panel (c) shows that higher view ratings are associated with progressively higher prices, suggesting a monotonic relationship between view quality and property value.

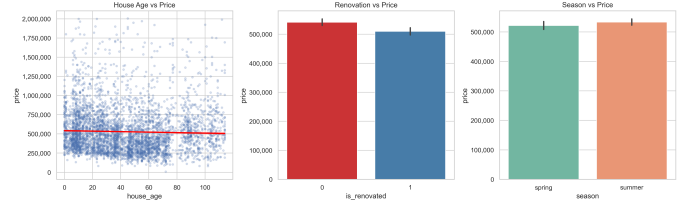


Fig. 3: Engineered features against price. House age, renovation status, and season individually carry weak signals but contribute non-additively to ensemble models.

These constructed features, as shown in Fig. 3, have very little predictive power by themselves. The correlation shown in Fig. 3a is a weak negative one between the age of the house and the price. Fig. 3b shows that there is only a

slight difference between the prices of houses that have been renovated and those that have not been. Fig. 3c shows that there is not much variation depending on the season.

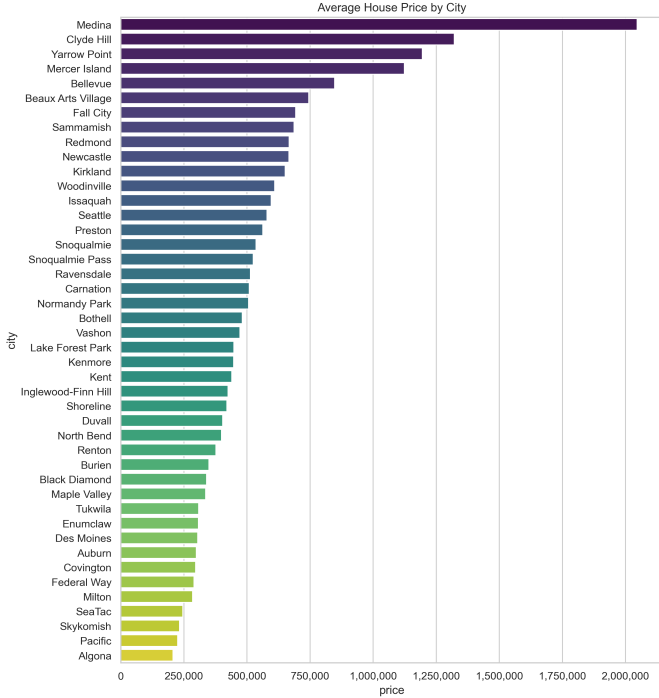


Fig. 4: Average house price by city. The distribution is highly skewed, with a small number of cities accounting for the high end of the market, motivating the use of target encoding.

City effects (Fig. 4) are sizable and heavy-tailed, as just a few cities are responsible for the higher end. With numerous cities included in the dataset, one-hot encoding would be redundant since it increases the dimensionality of the feature space without adding any value.

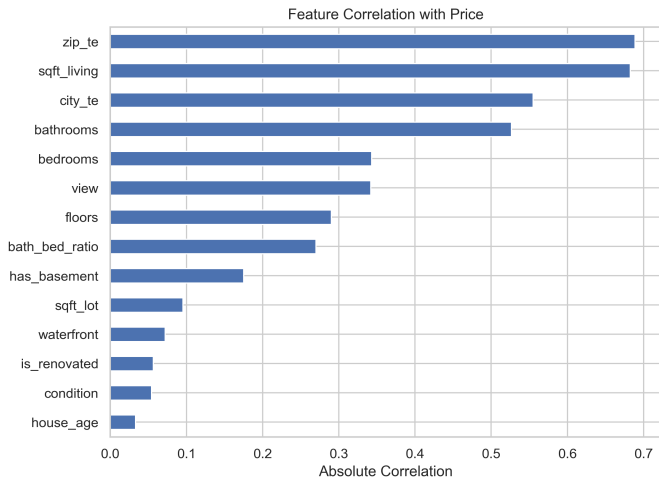


Fig. 5: Absolute correlation of each retained feature with price. *zip_te*, *sqft_living*, and *city_te* are the three strongest individual predictors.

The ranking of correlation shown in Figure 5 gives the inputs that have contributed the most to the signal. The top three predictors include *zip_te* ($r=0.689$), *sqft_living* ($r=0.683$), and *city_te* ($r=0.555$). The secondary set of predictors comprises *bathrooms* ($r=0.527$), *bedrooms* ($r=0.343$), and *view* ($r=0.342$). While the engineered features rank relatively low, they are kept according to the rules outlined above. The importance of both the location variables encoded by target encoding stands out as a common theme for the rest of the analysis.

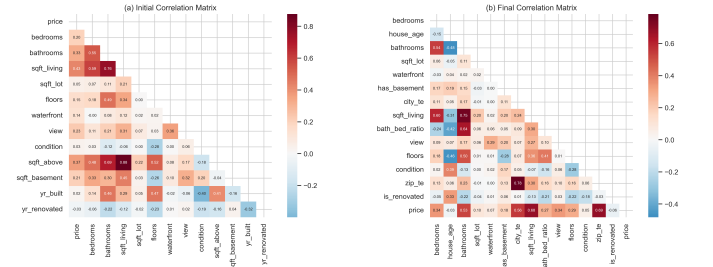


Fig. 6: Correlation matrix before and after feature selection. (a) Strong multicollinearity among size-related features. (b) After feature selection, redundant variables are removed while preserving relationships with the target.

The before-and-after comparison (Fig. 6) illustrates the effect of the multicollinearity reduction phase. In the initial correlation matrix, strong multicollinearity is observed among size-related features such as *sqft_living*, *sqft_above*, and *sqft_basement*, as well as between *bedrooms* and *bathrooms*. After feature selection, redundant variables are removed, resulting in a more compact 14-feature set. The reduced feature set mitigates multicollinearity while preserving the primary relationships with the target variable.

B. Model Comparison

TABLE I: Test-set performance of all models, sorted by R^2 .

Model	R^2	RMSE	MAE
Stacking V2	0.8015	124340	77588
Stacking V3	0.8003	124739	77899
Stacking V1	0.7867	128916	78842
LightGBM	0.7761	132083	81037
XGBoost	0.7651	135282	80934
Random Forest	0.7542	138377	83628
GBR	0.7539	138477	82641
Extra Trees	0.7336	144052	86086
Ridge	0.7080	150818	94766

The comparison between the models is shown in Table I and Figure 7. From the individual models, LightGBM is the strongest ($R^2 = 0.7761$), followed by XGBoost ($R^2 = 0.7651$). The top positions among individual models belong to LightGBM, XGBoost, GBR, Random Forest, and Extra Trees models, while Ridge is at the bottom of the list, which is consistent with exploratory data analysis findings that indicate highly important variables have non-linear relationships. Each of the three versions of stacking shows better results compared

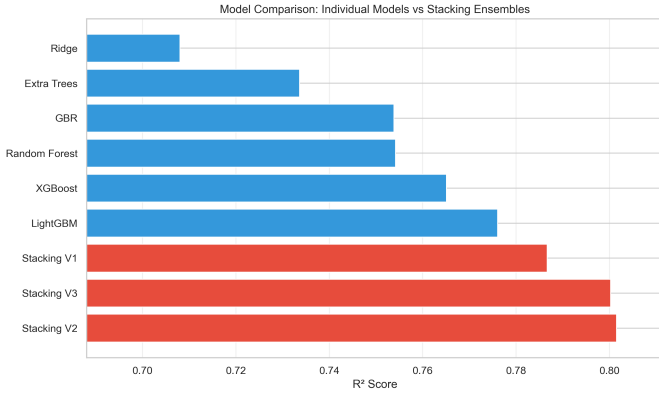


Fig. 7: Test R^2 for all nine models. Stacking variants narrowly beat the best individual model.

to LightGBM. The best results were achieved with Stacking V2 (Out-of-Fold predictions with passthrough + Ridge regression), with the highest R^2 value of 0.8015, which is 0.025 better compared to LightGBM and 0.036 better compared to XGBoost. Moreover, it should be noted that this superiority is demonstrated for all three versions of stacking and is far above any inter-model difference shown in the bottom part of the table, which means that the result obtained can be considered real and not random. At the same time, V2 of stacking is better by 0.015 than V1, showing a more informative feature space for the meta-learner.

C. Ablation and Meta-Learner Coefficients

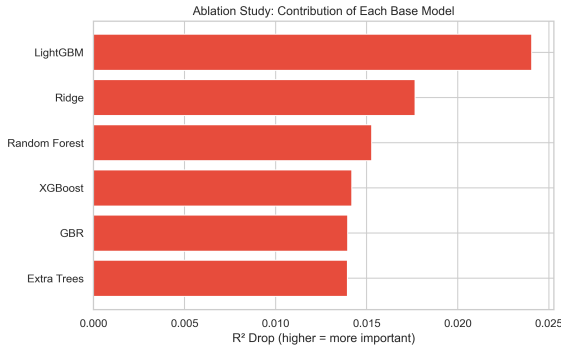


Fig. 8: Leave-one-out ablation on the best stacking model (V2). The performance drop (R^2 decrease) is measured after removing each base model; larger values indicate a greater contribution to the ensemble.

As shown in Fig. 8 and Table II, LightGBM is the most critical contributor to the overall performance. When omitted, there is a reduction of R-squared value of $R^2 = 0.0241$, which is 36% larger compared to the next highest reduction. Next comes Ridge, which produces a reduction of the R-squared value of $R^2 = 0.0177$. Notably, Ridge is the least performing model among all base models as shown in Table I. The other four models, Random Forest, XGBoost, GBR, and Extra Trees, are seen to make a similarly sized contribution of about R^2

TABLE II: Leave-one-out ablation on the best stacking model (V2).

Removed model	Resulting R^2	ΔR^2
LightGBM	0.7775	-0.0241
Ridge	0.7839	-0.0177
Random Forest	0.7863	-0.0153
XGBoost	0.7874	-0.0142
GBR	0.7876	-0.0140
Extra Trees	0.7876	-0.0139
None (full)	0.8015	0

TABLE III: Meta-learner coefficients on the OOF predictions. V2 uses Ridge; V3 uses LassoCV.

Base learner	V2 (Ridge)	V3 (Lasso)
LightGBM	+0.3672	+0.4163
Random Forest	+0.2333	+0.2305
Extra Trees	+0.1814	+0.0358
XGBoost	+0.1483	+0.1725
GBR	+0.0684	+0.0559
Ridge	-1.0206	0 (dropped)

= 0.0139-0.0153. This pattern is further supported by the meta-learner coefficients in Table III. In the V2 configuration (Ridge meta-learner), LightGBM receives the highest weight (+0.3672), followed by Random Forest and Extra Trees, while all tree-based models obtain positive contributions. Notably, Ridge is assigned a negative coefficient, suggesting that it acts as a corrective linear component rather than a primary predictor. For V3, the LassoCV meta-learner leads to a similar overall conclusion. LightGBM remains the dominant contributor, while less informative components are shrunk toward zero. In particular, the Ridge base learner is assigned a coefficient of zero, indicating that its contribution becomes redundant under L1 regularisation. Despite this simplification, V3 achieves a performance level very close to V2 ($R^2 = 0.8003$), suggesting that a reduced subset of models can still capture most of the predictive signal. This observation implies that the role of Ridge in V2 is not to provide strong standalone predictions, but rather to capture residual linear structure and correct systematic biases from tree-based models. Overall, these results consistently show that LightGBM provides the most essential predictive signal, while the other tree-based models contribute smaller but complementary information. The linear component plays a supporting role, improving robustness without dominating the final prediction.

D. Model Diagnostics

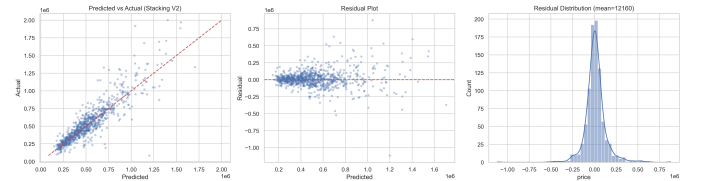


Fig. 9: Diagnostics for the best-performing stacking model (V2): predicted vs actual, residuals vs predicted, and residual distribution.

The results shown in Figure 9 confirm what is presented in the table. As can be seen from Fig. 9a, the predictions are tightly clustered around the line of identity. Fig. 9b shows that the residuals are centered around zero but their variance increases with increasing predicted values, which implies some degree of heteroscedasticity. This suggests reduced predictive accuracy for higher-priced properties, which is a common characteristic of housing price data with a long-tailed distribution.

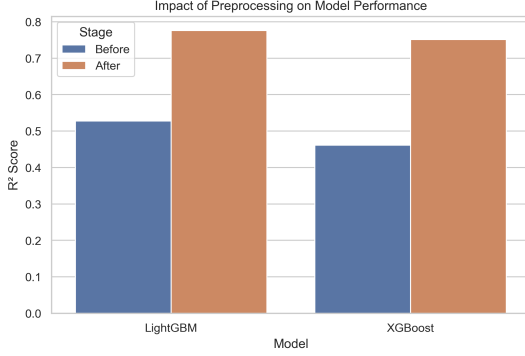


Fig. 10: R^2 for LightGBM and XGBoost before and after applying the preprocessing pipeline.

Figure 10 illustrates the impact of pre-processing. Without pre-processing, LightGBM achieves an R^2 score of 0.528, whereas XGBoost reaches an R^2 score of 0.461. After applying all the pre-processing steps in the pipeline, which involve removing any price zeroes, eliminating outliers, feature creation, target encoding, and logarithmic scaling of the target variable, the R^2 scores improve to 0.776 and 0.751, respectively. This improvement translates into a jump in R^2 of 0.248 and 0.290, respectively. These improvements are substantial and comparable in magnitude to the performance differences observed between models in Table I, indicating that preprocessing plays a critical role in improving predictive accuracy.

V. DISCUSSION

The hierarchy presented in Table I is more indicative of the nature of the data than the inherent merits of the algorithm families. The three boosting learners (LightGBM, XGBoost, GBR) score highest, followed by the two bagging learners (Random Forest, Extra Trees), and finally Ridge. This trend is consistent with the results of exploratory data analysis, according to which *sqft_living* is related to the house price non-linearly, waterfront and view entail discontinuous price changes, and the target-encoded location variables carry strong categorical information. Therefore, the boosting learners are well-suited to capture all three patterns, while Ridge is unable to fit non-linear relationships, putting it at a disadvantage from the start [16], [17]. Ensemble stacking brings about a significant improvement over the best individual model. All three stacking configurations outperform LightGBM by 0.011 to 0.025 in R^2 consistently across the three setups, and

the gains are larger than the differences between individual models within the same performance band. The ablation study reveals the mechanism behind this boost. LightGBM accounts for most of the predictive power ($\Delta R^2 = 0.0241$), whereas the other four tree-based learners (Random Forest, XGBoost, GBR, Extra Trees) provide contributions ranging from 0.0139 to 0.0153 each, approximately equal but small individually. Thus, the meta-learner does not pick a single dominant model and dismiss the others; rather, it combines the main prediction with multiple smaller corrections from other tree-based learners. This result agrees with the finding by Bjørge et al. [10] that stacking benefits from having diverse base learners, and indicates that employing only one single best learner is insufficient to exploit this benefit fully.

The ablation analysis leads to another important insight. Ridge ranks last in terms of standalone performance in Table I ($R^2 = 0.7080$); however, it incurs the second-largest loss upon removal from the ensemble ($\Delta R^2 = 0.0177$), behind LightGBM. This apparent contradiction arises from the fundamentally different types of signal captured by Ridge and the tree-based learners. While the latter produce complex non-linear patterns through interactions between multiple features, Ridge captures a simple additive linear structure over standardised features, particularly the target-encoded spatial variables. Although tree-based learners can approximate linear relationships, they do so less efficiently and require many splits to represent smooth trends, making them less effective at capturing globally consistent linear signals. As a result, the meta-learner cannot fully reconstruct this linear component solely from its predictions. The behaviour of the Lasso meta-learner in V3 further supports this explanation. When the original features are included via passthrough, the meta-learner assigns a zero coefficient to Ridge, effectively removing it from the ensemble. At the same time, the linear structure is captured directly from the raw input features, particularly the spatial variables. Despite this change in representation, V3 achieves a performance level very close to that of V2 ($R^2 \approx 0.8003$ vs. 0.8015), indicating that the linear signal remains present in the model, but is expressed through a different pathway. This suggests that the contribution of a base learner in an ensemble depends not only on its standalone predictive accuracy, but also on the uniqueness of the signal it provides. In this case, Ridge contributes a complementary linear component, which can either be explicitly modelled as a base learner or implicitly recovered from the original features when available.

Finally, two important observations run throughout the model comparison process. Firstly, the preprocessing experiment (Fig. 10) shows that data preparation is equally important as model choice: LightGBM improves from $R^2 = 0.528$ to 0.776 with the complete data pipeline (zero-price filter, outlier removal, feature engineering, target encoding, log-transformed target), and XGBoost exhibits a similar gain from 0.461 to 0.751. The increase amounts to roughly 0.25 to 0.29, similar to the difference between the best and worst models in Table I. This suggests that preprocessing should not be

treated as a standardised procedure to be done preliminarily before modelling but rather take it as a modelling decision in itself. Secondly, the strong ranking of the target-encoded location features (*zip_te* ($r = 0.689$) and *city_te* ($r = 0.555$)), alongside *sqft_living* ($r = 0.683$), highlights the importance of spatial information in housing valuation. It aligns with the long-recognised limitation of hedonic pricing models in accounting for spatial heterogeneity [13] and proves that target encoding is a low-cost means of recovering this information without suffering from the dimensionality of one-hot encoding.

VI. CONCLUSION AND FUTURE WORK

This paper introduces a stacking model for predicting house prices based on a controlled experiment and an ablation analysis. The optimal stacking model achieves $R^2 = 0.8015$ on King County test data, demonstrating a gain of 0.025 compared to the best-performing base model (LightGBM, $R^2 = 0.7761$) and 0.036 compared to XGBoost. Both ablation and meta-learner coefficient results converge on a common core principle: LightGBM is responsible for most of the signal; Random Forest, XGBoost, GBR, and Extra Trees add corrections of a similar order; Ridge, though the weakest base model, is the second-best base model because it contributes an important linear signal that cannot be added by tree-based algorithms. Finally, the Lasso meta-learner demonstrates the same conclusion with a slightly different methodology: it eliminates Ridge from the OOF block and brings back the linear signal by using passthroughs, with virtually no drop in the accuracy. In practice, the utility of a base learner in a stack lies in its contribution to the predictive signal, not in its standalone ability.

In addition to the primary finding, there are two side results in this paper. First, the impact of preprocessing techniques is comparable to the impact of model selection: the performance of LightGBM increases from $R^2 = 0.528$ to 0.776, and that of XGBoost increases from 0.461 to 0.751 when all preprocessing steps are accounted for, with a log-transform of the target used to handle the heavy upper tail of the price distribution. Second, features obtained by target encoding zip codes and cities turn out to be highly predictive, in a company with *sqft_living* feature, which proves that geography plays a critical role in price variation, and that target encoding manages to capture this signal without the expense of increasing dimensionality incurred by one-hot encoding.

The following three problems are identified in this research. First, the model fails on luxury homes because both the 99th percentile cutoff and the chosen features represent luxury homes poorly. Second, all features are tabular; macroeconomic, schooling, or geographic information may provide additional information for the price estimation problem. Third, stacking does not seem to offer much value beyond the inclusion of LightGBM as a base learner. Three research directions are suggested for future work. First, train models using specialized loss functions or explicit sample weighting to compensate for heteroscedasticity in price data. Second, use learned embeddings or coordinates as features to encode

wealth information geographically. Third, refine the approach to ablation by choosing base learners based on their diversity prior to meta-learner training.

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